Ain Shams University
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Thermal Management of a Battery-Powered Medical Payload Pod

A scaled prototype study inspired by drone-delivered emergency medical care

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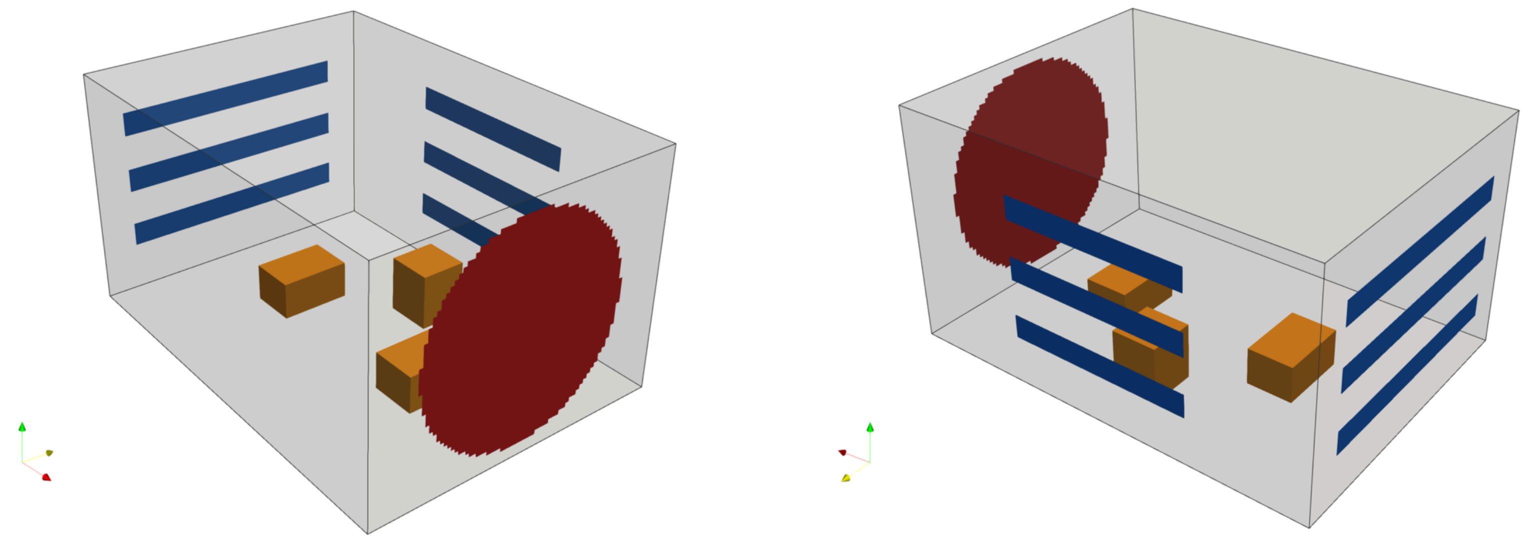
Heat Transfer · MEP 311s · Presented to Dr. Amr Elbanhawy · 16 June 2026

The Problem

- Drone-delivered medical payload pod (Zipline-style): **three** thermally managed enclosures — power, control, plant
- The **plant module** dissipates **~17 W** (two 8 Ω resistors + IRF530 MOSFET)
- **Constraint:** every module must stay **≤ 40 °C** in steady state
- Judged on thermal performance **and** cost-to-weight ratio

Design — Forced-Air Fan + Vents



Fan-wall view: red disk = 45 mm exhaust fan; orange = heat sources

Vent-wall view: blue slots = 6 intake vents (3 front + 3 side)

- One **50 mm 5 V exhaust fan** (45 mm opening) + **six intake slots**, in a 90 × 65 × 50 mm PLA box
- Selected over a **fin array** (≈ 0.30 m^2 of fins, far heavier & bulkier)

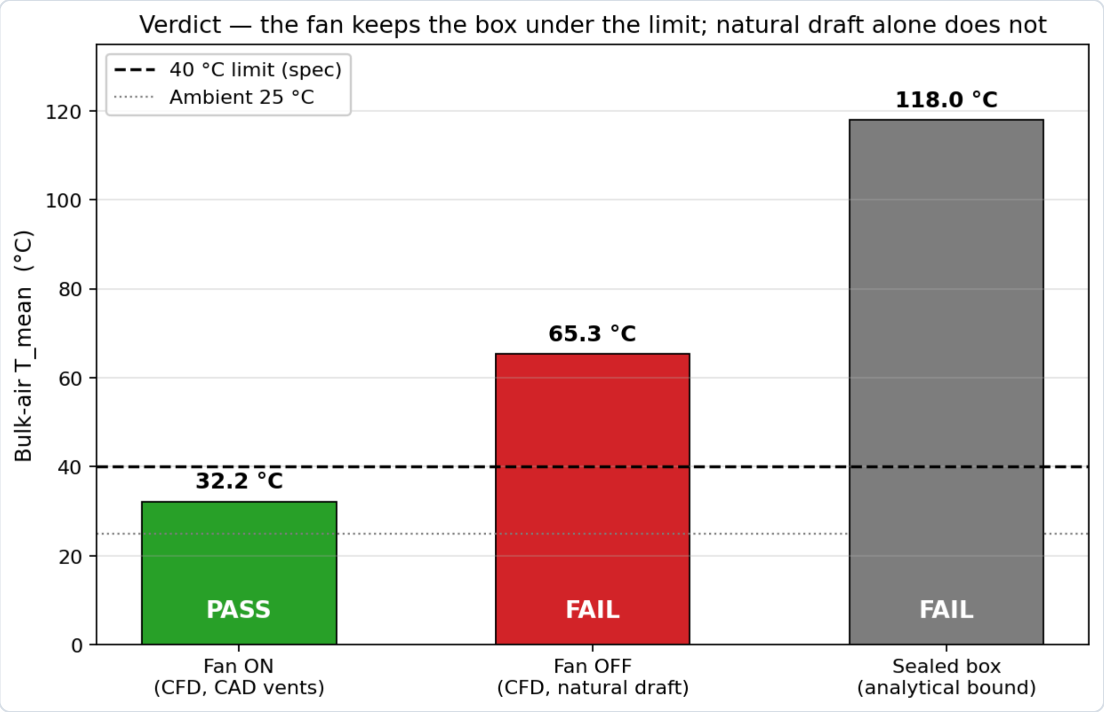
Analytical Thermal Analysis

- **Power:** $P = V^2/R = 8^2/4 =$ **16 W** resistive (~17 W total)
- **Fin array — rejected:** needs ~0.30 m^2 over two faces; heavy and hard to mount
- **Forced-air energy balance:** $\Delta T = Q/(\rho \cdot \dot{V} \cdot c_p) \approx$ **4.6 °C** → cavity ~30–35 °C
- Bracketed by the sealed-box bound (118 °C); confirmed by CFD & experiment

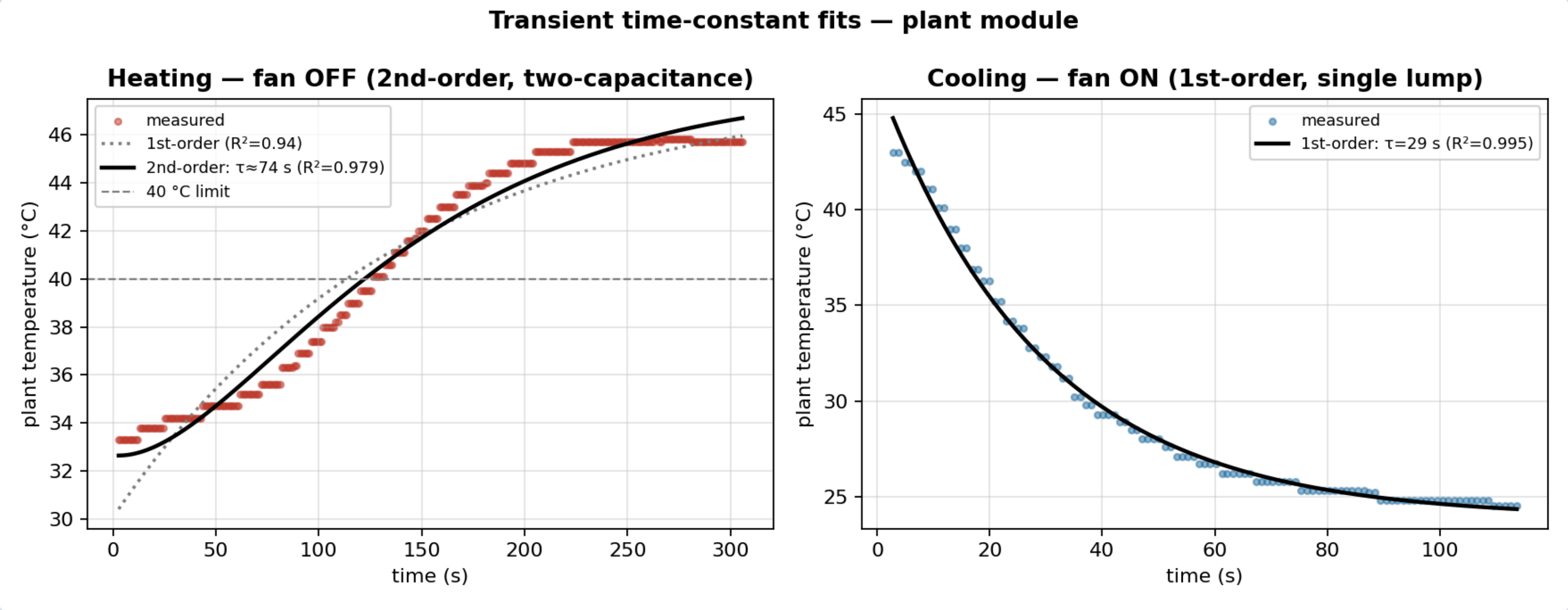
CFD Simulation BONUS

- OpenFOAM · k-ω SST · **528 k cells**
- **Fan ON → 32 °C → PASS**
- Fan OFF → 65 °C → FAIL
- Heat hugs the floor; ~¾ leaves as warm fan exhaust — energy balance closes on 17 W

32 °C	65 °C
Fan ON (PASS)	Fan OFF (FAIL)

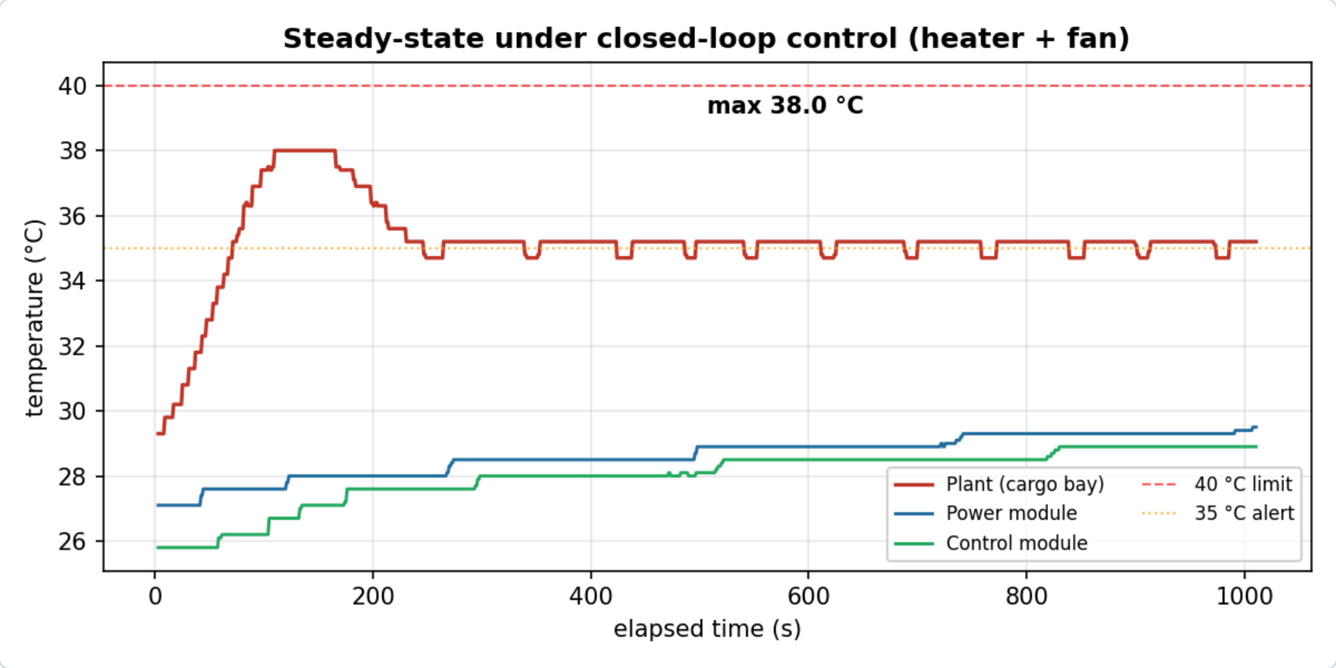


Transient Response



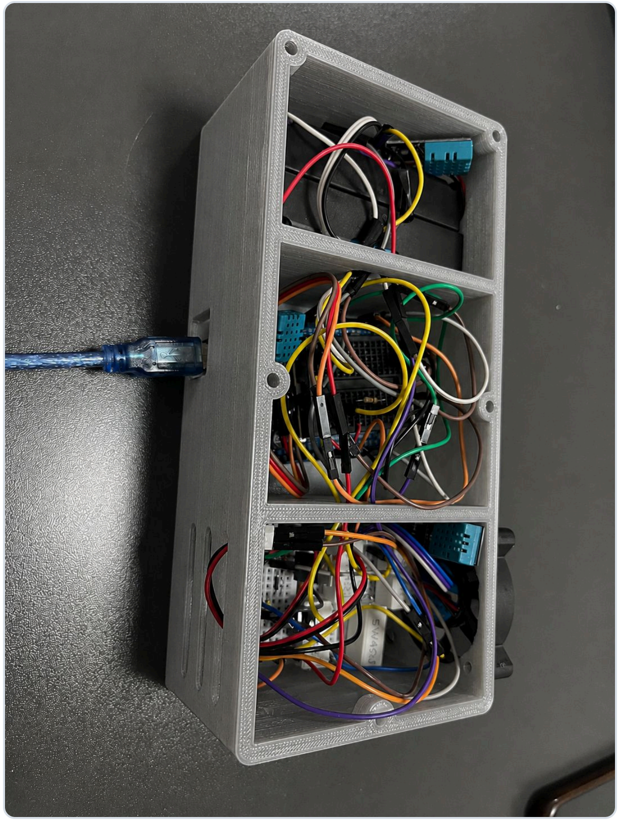
- **Heating (fan off): second-order** — two cascaded thermal capacitances (S-shaped)
- **Cooling (fan on):** clean **first-order, τ = 29 s** (R^2 = 0.995); predicted C/UA ≈ 20–30 s

Experimental Results



- Closed-loop control holds the plant at **~35 °C** (< 40 °C); power & control ~29 °C
- Measured full-heat draw **17.1 W** — matches the 17 W used in the analysis

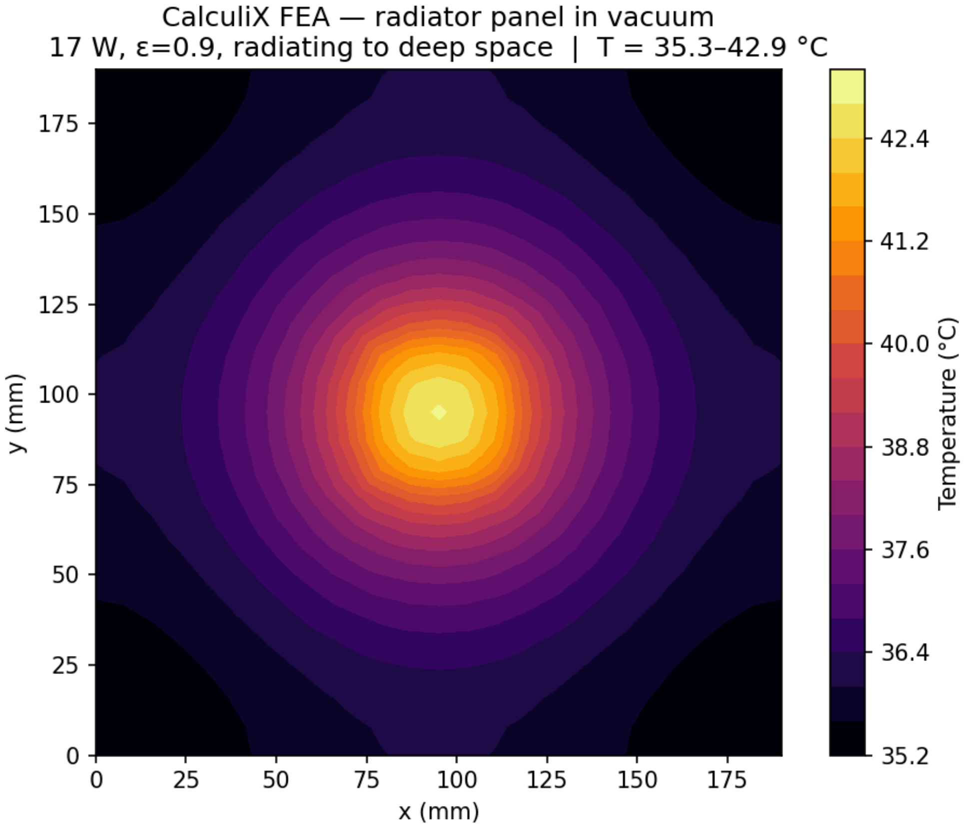
Prototype & Cost



- 3-D-printed three-compartment PLA pod
- Arduino **PWM** control of the MOSFET; 3× DHT sensors; graded LED alert + 40 °C cutoff
- Battery + load **current/power** monitored in firmware
- Thermal-management cost-to-weight: **80 EGP / 55 g ≈ 1.45 EGP/g**

Space-Environment Variant BONUS

- In vacuum there is **no convection** — the fan/vents do nothing
- Heat is rejected by **radiation** to deep space via a high-emissivity panel
- Sizing: a **~0.035 m²** radiator holds the module at 40 °C
- **CalculiX FEA:** panel 35–43 °C (mean ~37 °C) — confirms the sizing



Conclusion

- **All modules ≤ 40 °C** in steady state — constraint met and experimentally confirmed
- The fan is mandatory (removes 33 °C); the vents are a passive safety net
- Two analytical methods + CFD + experiment all agree (~32–37 °C)
- Bonuses delivered: **numerical** (CFD + CalculiX FEA) and the **space** variant
- Code & data: **github.com/omarkeshk/medical-payload-pod-thermal**